ASSIGNMENT 4:

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1. Create a class named ***'Rectangle'*** with two data members- *length* and *breadth* and a function to calculate the area which is 'length*breadth'. The class has *three constructors* which are :

(a) having no parameter - values of both length and breadth are assigned zero.

(b) having two numbers as parameters - the two numbers are assigned as length and breadth respectively.

(c) having one number as parameter - both length and breadth are assigned that number.

Now, create objects of the 'Rectangle' class having none, one and two parameters and print their areas.

C++

```cpp
#include <iostream>
using namespace std;
class Rectangle {
private:
    int length, breadth;
public:
    // Default constructor
    Rectangle() {
        length = 0;
        breadth = 0;
    }
    // Two parameter constructor
    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }
    // One parameter constructor
    Rectangle(int x) {
        length = breadth = x;
    }
    int area() {
        return length * breadth;
    }
};
int main() {
    Rectangle r1;
    Rectangle r2(10);
    Rectangle r3(5, 4);
    cout << "Area r1 = " << r1.area() << endl;
    cout << "Area r2 = " << r2.area() << endl;
    cout << "Area r3 = " << r3.area() << endl;
    return 0;
}
```

Run | Visualize Code

Enter Input here

If your code takes input, add it in the abov

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 3.616 Mb

Your Output

```
Area r1 = 0
Area r2 = 100
Area r3 = 20
```

print their areas.

2. Redefine the above program by creating an array of objects of the class Rectangle and calculate area for each object calling different constructors. Also implement constructors with default arguments and destructor in this program.

```cpp
#include <iostream>
using namespace std;

class Rectangle {
private:
    int length, breadth;

public:
    Rectangle() {
        length = 0;
        breadth = 0;
    }

    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }

    Rectangle(int x) {
        length = breadth = x;
    }

    int area() {
        return length * breadth;
    }
};

int main() {
    Rectangle arr[3] = {
        Rectangle(),
        Rectangle(7),
        Rectangle(6, 3)
    };

    for (int i = 0; i < 3; i++) {
        cout << "Area arr[" << i << "] = " << arr[i].area() << endl;
    }

    return 0;
}
```

Enter Input here

If your code takes input, add it in the ab

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 3.532 Mb

Your Output

```
Area arr[0] = 0
Area arr[1] = 49
Area arr[2] = 18
```

3. Verify the following about *destructor* by writing the program:
   - (i) Name should begin with tilde sign(~) and must match class name.
   - (ii) There cannot be more than one destructor in a class.
   - (iii) Destructors do not allow any parameter.
   - (iv) They do not have any return type, just like constructors.

When you do not specify any destructor in a class, compiler generates a default destructor and inserts it into your code.

```cpp
#include <iostream>
using namespace std;

class Rectangle {
private:
    int length, breadth;

public:
    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }

    int area() {
        return length * breadth;
    }

    // Destructor
    ~Rectangle() {
        cout << "Destructor called\n";
    }
};

int main() {
    Rectangle r1(5, 4);
    cout << "Area = " << r1.area() << endl;

    return 0;
}
```

Enter Input here

If your code takes input, add it in the ab

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 3.512 Mb

Your Output

```
Area = 20
Destructor called
```

destructor and inserts it into your code.

4. Implement dynamic memory allocation. Use *new* and *delete* keywords.
(For integer variable, float variable, integer array, float array, class objects, Array of Objects)

C++

```cpp
#include <iostream>
using namespace std;
class Rectangle {
private:
    int length, breadth;
public:
    Rectangle() {
        length = 0;
        breadth = 0;
    }
    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }
    int area() {
        return length * breadth;
    }
};
int main() {
    Rectangle *ptr = new Rectangle(8, 2);
    cout << "Area = " << ptr->area() << endl;
    delete ptr;
    Rectangle *arr = new Rectangle[2];
    cout << "Area arr[0] = " << arr[0].area() << endl;
    cout << "Area arr[1] = " << arr[1].area() << endl;
    delete[] arr;
    return 0;
}
```

Run | Visualize Code

Enter Input here

If your code takes input, add it in the abov

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 3.484 Mb

Your Output

```
Area = 16
Area arr[0] = 0
Area arr[1] = 0
```